

Maximum Power Transfer Theorem

Actual Statement of Theorem

The Maximum Power Transfer Theorem states that maximum power is delivered from a source to a load when the load resistance is equal to the Thevenin (or internal) resistance of the source network.

$$R_L = R_{th}$$

This condition ensures that the power transferred to the load is at its highest possible value.

Mathematical Explanation

From Thevenin's equivalent circuit, the current through the load is:

$$I = V_{th} / (R_{th} + R_L)$$

Power delivered to the load is:

$$P = I^2 \times R_L$$

Substituting current:

$$P = (V_{th}^2 \times R_L) / (R_{th} + R_L)^2$$

By differentiating this expression with respect to R_L and setting it to zero, the condition for maximum power is obtained:

$$R_L = R_{th}$$

Maximum Power Formula

When $R_L = R_{th}$, the maximum power delivered is:

$$P_{max} = V_{th}^2 / (4 \times R_{th})$$

This shows that only half of the total power generated is actually delivered to the load.

Step-by-Step Procedure

Step 1: Convert Circuit to Thevenin Equivalent

Simplify the circuit into a single voltage source (V_{th}) and resistance (R_{th}).

Step 2: Identify Load Resistance

Determine the load resistor R_L connected to the circuit.

Step 3: Apply Maximum Power Condition

Adjust R_L such that:

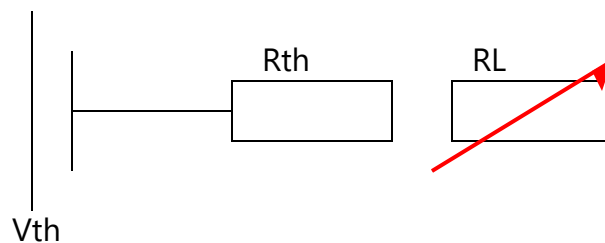
$$R_L = R_{th}$$

Step 4: Calculate Maximum Power

Use the formula:

$$P_{max} = V_{th}^2 / (4R_{th})$$

Circuit Diagram



The arrow across R_L indicates a **variable resistor**. By adjusting R_L , maximum power transfer can be achieved when R_L equals R_{th} .

Relation with Norton Theorem

The theorem also applies to Norton equivalent circuits:

$$R_L = R_N$$

Thus, maximum power transfer condition remains the same regardless of representation.

Important Observations

Although maximum power is transferred when $R_L = R_{th}$, this condition is not always efficient. Only 50% of the power is delivered to the load, and the remaining is lost in the internal resistance.

Therefore, in power systems, efficiency is often prioritized over maximum power transfer.

Practical Applications

This theorem is widely used in communication systems where maximum signal strength is required, such as in antenna and amplifier design.

It is also used in audio systems for impedance matching to ensure maximum signal transfer.

However, in power transmission systems, engineers avoid this condition to reduce energy loss and improve efficiency.